

The MX Model

Importable Goods, Exportable Goods, and the Terms of Trade

The MX Model: Why Do We Need It?

- So far, the empirical SVAR has shown how the economy **responds to Terms-of-Trade shocks**.
- But the previous one-good framework treats the ToT shock essentially as a productivity shock.
- This implies an unrealistic economy in which:
 - virtually all domestic production is exportable;
 - virtually all domestic absorption is imported.
- In the data, however, countries both **produce and consume** importable and exportable goods.

Key Innovation of the MX Model

The economy explicitly contains two sectors: **Importables (m)** and **Exportables (x)**.

Empirical SVAR $\implies$ Observed responses

MX Model $\implies$ Structural mechanism

The MX Model: Households

Households choose consumption, sector-specific labor, investment, capital accumulation, and external debt:

$$\max E_0 \sum_{t=0}^{\infty} \beta^t U(c_t, h_t^m, h_t^x)$$

subject to

$$\begin{aligned} c_t + i_t^m + i_t^x + \Phi_m(k_{t+1}^m - k_t^m) + \Phi_x(k_{t+1}^x - k_t^x) + d_t \\ = \frac{d_{t+1}}{1 + r_t} + w_t^m h_t^m + w_t^x h_t^x + u_t^m k_t^m + u_t^x k_t^x. \end{aligned}$$

Capital evolves separately in each sector:

$$k_{t+1}^j = (1 - \delta)k_t^j + i_t^j, \quad j \in \{m, x\}.$$

- Labor and capital are **sector-specific**.
- Adjustment costs prevent instantaneous reallocation across sectors.
- Households can borrow or lend internationally.

The MX Model: Production Structure

Sectoral production

$$y_t^m = F^m(k_t^m, h_t^m), \quad y_t^x = F^x(k_t^x, h_t^x).$$

Firms maximize profits:

$$p_t^j F^j(k_t^j, h_t^j) - w_t^j h_t^j - u_t^j k_t^j,$$

implying

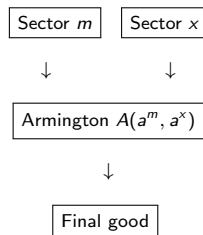
$$p_t^j F_1^j = u_t^j, \quad p_t^j F_2^j = w_t^j.$$

Final-good production

$$A(a_t^m, a_t^x)$$

combines importable and exportable goods
using an **Armington aggregator**.

$$A_1(a_t^m, a_t^x) = p_t^m, \quad A_2(a_t^m, a_t^x) = p_t^x.$$



Market Clearing and the Terms-of-Trade Shock

Domestic final-good market clearing:

$$c_t + i_t^m + i_t^x + \Phi_m(\Delta k_t^m) + \Phi_x(\Delta k_t^x) = A(a_t^m, a_t^x).$$

Imports and exports are

$$m_t = p_t^m(a_t^m - y_t^m), \quad x_t = p_t^x(y_t^x - a_t^x).$$

External debt evolves according to

$$\frac{d_{t+1}}{1 + r_t} = d_t + m_t - x_t.$$

Terms of Trade

$$tot_t = \frac{p_t^x}{p_t^m}, \quad \widehat{tot}_t = \rho \widehat{tot}_{t-1} + \pi \varepsilon_t^{tot}.$$

- As in the SVAR, the economy is small and takes the ToT as **exogenous**.
- Importantly, the same estimated ρ and π from the empirical exercise discipline the model.

From the Model to the Data

- With multiple goods, model variables cannot be compared directly with the empirical aggregates used in the SVAR.
- The WDI measures real GDP using a **Paasche implicit price deflator**.
- Therefore, the model constructs theoretical counterparts consistent with the empirical measures.

Nominal GDP is

$$P_t^m y_t^m + P_t^x y_t^x.$$

Using the Paasche deflator, model-consistent real GDP becomes

$$GDP_t^{real} = p^m y_t^m + p^x y_t^x$$

where steady-state relative prices are used as base-period prices.

Why This Matters

The MX model and the empirical SVAR must measure **the same economic objects** before their predictions can be compared.

Functional Forms and Economic Frictions

Preferences: GHH

$$U(c, h^m, h^x) = \frac{[c - G(h^m, h^x)]^{1-\sigma} - 1}{1-\sigma}.$$

Sectoral labor

$$G(h^m, h^x) = \frac{(h^m)^{\omega_m}}{\omega_m} + \frac{(h^x)^{\omega_x}}{\omega_x}.$$

Production: Cobb–Douglas

$$F^j(k^j, h^j) = A^j (k^j)^{\alpha_j} (h^j)^{1-\alpha_j}.$$

Capital adjustment costs

$$\Phi_j(z) = \frac{\phi_j}{2} z^2.$$

Debt-elastic premium

$$p(d) = \bar{p} + \psi(e^{d-\bar{d}} - 1).$$

Armington aggregator

$$A(a^m, a^x) = [\chi(a^m)^{1-1/\mu} + (1-\chi)(a^x)^{1-1/\mu}]^{\frac{1}{1-1/\mu}}.$$

- ϕ_m, ϕ_x : slow capital reallocation.
- ω_m, ω_x : imperfect labor mobility.
- μ : substitution between importables and exportables.
- ψ : stabilizes external debt.

Parameterization of the MX Model

The model combines **calibration** and **country-level estimation**.

1 Parameters inherited from standard SOE models

$$\sigma, \delta, r^*, \bar{p}, \alpha_m, \alpha_x, \omega_m, \omega_x.$$

2 Parameters disciplined by sectoral trade data

$$\mu, \chi, \bar{d}, A_x.$$

They match average trade-balance, export, and exportable-output shares.

3 Country-specific parameters

$$\rho, \pi, \phi_m, \phi_x, \psi.$$

Here, ρ and π come directly from the **estimated ToT process**.

	ρ	π	ϕ_m	ϕ_x	ψ
Median	0.53	0.09	1.82	1.56	0.18

Connection to the Empirical Exercise

The estimated ToT process from the first activity **feeds directly into the structural MX model**.

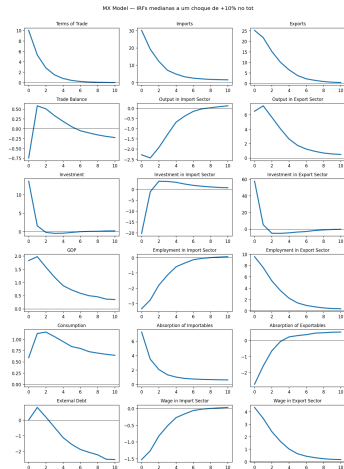
What Does the MX Model Predict?

Following a positive ToT shock:

- production shifts from **importables to exportables**;
- imports rise as domestic demand for importables increases;
- exports rise as exportable production expands;
- GDP and consumption increase;
- the trade balance **deteriorates on impact**.

Key Tension

The model fails to reproduce the empirical **HLM effect**: in the SVAR, the trade balance improves on impact.



MX model, cross-country median IRFs to a +10% ToT shock (replication).

Variance Explained by ToT Shocks: MX vs. SVAR

Cross-country median of the fraction of each variable's variance explained by terms-of-trade shocks (Table 7.6):

Variance share (%)	MX Model	Empirical SVAR
Terms of Trade	100.0	100.0
Trade Balance	26.9	12.2
Output	18.0	9.8
Consumption	24.1	10.6
Investment	20.3	10.2

Main Result

The MX model attributes roughly **twice as much** business-cycle variation to ToT shocks as the empirical SVAR.

Country-Level Variance Shares (Table 7.7)

Fraction of each variable's variance explained by ToT shocks (%), country by country: MX model vs. empirical SVAR. Ratios can exceed 100 when the model predicts more ToT-driven variance than the data display.

Country	Trade Bal.		Output		Consump.		Invest.	
	MX	SVAR	MX	SVAR	MX	SVAR	MX	SVAR
Argentina	4.1	25.3	5.0	30.5	5.4	17.0	5.0	25.9
Bolivia	12.5	2.5	17.9	3.5	59.6	10.6	45.1	7.6
Botswana	9.7	27.7	16.8	48.6	6.0	36.8	11.1	39.0
Brazil	60.4	43.4	15.2	10.9	14.4	3.7	19.6	16.2
El Salvador	20.0	10.7	19.9	10.3	22.3	10.2	23.6	10.7
Guatemala	39.1	5.4	18.1	2.5	12.1	3.4	30.4	4.8
Uruguay	4.1	17.3	8.3	34.3	6.2	32.9	3.8	14.4
Venezuela	26.9	41.9	15.1	23.8	9.5	33.6	8.3	11.0
Zimbabwe	3.2	7.0	4.7	10.3	7.2	33.8	2.2	16.9
Median (all 51)	26.9	12.2	18.0	9.8	24.1	10.6	20.3	10.2

Reading the Table

Considerable heterogeneity across countries; the median (Table 7.6) is the robust summary. Full 51-country table in `table_7.7_bycountry.csv`.